
PHYSICS GROUP PROJECT REPORT

Rotational and Frictional Dynamics of the Slamming of a Door

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Project Group: Team solvE (Group 14)
Group Members: Eunwoo Chae (*Data analysis, GitHub, Report $\LaTeX$*)
Seongmin Kang (*Report & computation*)
Minseok Jang (*Experiment*)
Solbi Joo (*Presentation & video*)
Course Title: General Physics
Instructor: Prof. Deniz Olgu Devecioglu
Department: DGIST

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Abstract

The slamming of a door is a simple instance of damped rotational motion in which more than one friction mechanism may act simultaneously. Klein et al. examined which of the three standard velocity dependences of the frictional torque, constant (dry), linear (Stokes), or quadratic (Newtonian air drag), best describes the motion. We reproduce their analysis both theoretically and experimentally. The angular velocity of a real door was measured directly with a smart-phone gyroscope, so that the conversion from radial acceleration used in the original work was not required. Two slams of different initial spin were recorded, the six friction models were fitted to the free-rotation portion of each, and the equation of motion was integrated numerically to verify the closed-form solutions. Following the original paper, the models are distinguished not by goodness of fit but by the physical plausibility of the fitted coefficients. On this basis the Stokes and dry-friction models are rejected, and the quadratic air-drag term is found to be the only physically consistent description, in agreement with Klein et al.

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1 Introduction and Contextual Background

1.1 Historical and Pedagogical Significance

Damped rotational motion is commonly introduced with a single friction term chosen for analytical convenience, a constant torque in one treatment and a term linear in velocity in another. Klein et al. point out that this practice obscures the question of which mechanism actually dominates a given system, and they adopt a familiar example for which students already possess physical intuition, the closing of a door. The paper is aimed at introductory and lower-division undergraduates, and its principal contribution is methodological. A friction model may reproduce the data closely and still be incorrect, so the choice between candidate models must rest on a comparison of the fitted coefficients with independent physical estimates. This principle guides the present report.

1.2 Physical Setup and System Parameters

The door rotates about a vertical hinge axis, so that gravity produces no torque and the motion is opposed only by friction at the hinges and by air drag on the door surface. We model the door as a uniform thin rectangular plate of width w and mass m , with a smartphone fixed at a distance r from the axis. In contrast to the original experiment, which recorded a radial acceleration and converted it through $\omega = \sqrt{a_r}/r$, the gyroscope used here reports the angular velocity directly and removes this intermediate step. The measured quantities are listed in Table 1.

Table 1: Symbols and measured values for the door.

Symbol	Meaning	Value / unit
m	mass of the door	13.7 kg
w	width	0.833 m
h	height	2.031 m
r	hinge-to-phone distance	0.759 m
ω	angular velocity	rad/s
τ_f	frictional torque	N m
a, b, c	dry, Stokes, Newton coefficients	N m, N m s, N m s ²
$I = \frac{1}{3}mw^2$	moment of inertia	3.17 kg m ²

2 Step-by-Step Mathematical Derivations

Several intermediate steps are stated without derivation in the original paper. We reproduce them explicitly for the three results on which the analysis depends, namely the moment of inertia, the form of the frictional torque, and the solution of the equation of motion.

2.1 Equation of Motion

For rotation about a fixed axis under a single net frictional torque opposing the motion, Newton's second law gives $-I d\omega/dt = \tau_f$. Taking the frictional torque to contain constant, linear, and quadratic contributions, $\tau_f = a + b\omega + c\omega^2$ with $a, b, c \geq 0$, yields

$$\frac{d\omega}{dt} = -(A + B\omega + C\omega^2), \quad A \equiv \frac{a}{I}, \quad B \equiv \frac{b}{I}, \quad C \equiv \frac{c}{I}. \quad (1)$$

The reduced coefficients A , B , and C are used throughout, since they are the quantities returned directly by a fit without separate knowledge of I .

2.2 Moment of Inertia of the Door

The moment of inertia about the hinge is obtained from $I = \int r_{\perp}^2 dm$ with $r_{\perp}^2 = x^2 + y^2$ for a plate of uniform density ρ .

Reducing I to $\frac{1}{3}mw^2$

Taking the hinge as the z -axis, width along x ($0 \leq x \leq w$) and thickness along y ($0 \leq y \leq d$),

$$I = \rho \int_0^h \int_0^d \int_0^w (x^2 + y^2) dx dy dz = \rho h d \left(\frac{1}{3}w^3 + w \frac{1}{3}d^2 \right) = \frac{1}{3}\rho h d w (w^2 + d^2).$$

A door is thin, $d \ll w$, so the d^2 term is negligible. Using $\rho = m/(whd)$,

$$I \approx \frac{1}{3}\rho h d w^3 = \frac{1}{3}mw^2.$$

For our door this is $I = \frac{1}{3}(13.7)(0.833)^2 = 3.17 \text{ kg m}^2$.

2.3 Origin of the Three Torque Terms

The three contributions to τ_f have distinct physical origins, which is what later allows the fitted coefficients to be judged. Dry friction at the bearings is essentially independent of speed, so $\tau_{\text{dry}} = a$. A Stokes drag force is proportional to the local velocity $v = r\omega$, and because torque introduces a further factor of r , it gives $\tau_{\text{Stokes}} = b\omega$. Newtonian air drag scales as v^2 , and the same argument gives $\tau_{\text{Newton}} = c\omega^2$. Which contribution dominates cannot be decided a priori.

2.4 Solving the Equation of Motion

Equation (1) is separable. Retaining a single term at a time gives the three one-mechanism solutions,

$$\omega_D(t) = \omega_0 - At, \quad \omega_S(t) = \omega_0 e^{-Bt}, \quad \omega_N(t) = \frac{\omega_0}{1 + C\omega_0 t}, \quad (2)$$

while the pairwise combinations give

$$\omega_{DS}(t) = \left(\omega_0 + \frac{A}{B} \right) e^{-Bt} - \frac{A}{B}, \quad (3)$$

$$\omega_{SN}(t) = \frac{B\omega_0}{(B + C\omega_0) e^{Bt} - C\omega_0}, \quad (4)$$

$$\omega_{DN}(t) = \sqrt{\frac{A}{C}} \tan\left(\arctan(\omega_0 \sqrt{\frac{C}{A}}) - \sqrt{AC} t\right). \quad (5)$$

The general solution for $a, b, c \neq 0$ is given in the appendix of the original paper. It is not needed here, since in every fit at least one coefficient was driven to zero. All six expressions are implemented in `src/friction_models.py`.

3 Physical Approximations and Limiting Cases

It is useful to ask in advance how distinguishable these solutions are. Expanding each for small t ,

$$\omega(t) \approx \omega_0 - (A + B\omega_0 + C\omega_0^2)t + \mathcal{O}(t^2), \quad (6)$$

so that to first order every model reduces to a straight line with the same initial slope. The curvature that distinguishes the dry, Stokes, and Newtonian cases appears only after ω has changed appreciably. A door slammed hard reaches the frame within a fraction of a second and loses little of its angular velocity beforehand, so its free-rotation record is short and nearly linear, and such a record cannot by itself separate the models.

A nearly linear $\omega(t)$ suggests a constant torque, and hence dry friction. Klein et al. caution against this inference, noting that over a limited range the data “look quite linear” even though this appearance does not constitute evidence for the dry model. The resolution is to set the fit quality aside and ask whether the fitted coefficient is physically reasonable, the approach taken in Section 5.

The two drag laws also correspond to different flow regimes. Stokes drag applies at low Reynolds number and Newtonian drag at high Reynolds number. A door moving through air at of order 1 rad/s lies well within the high-Reynolds regime, which already indicates which term is expected to dominate physically.

4 Computational and Experimental Verification

4.1 Measurement

A smartphone (Samsung SM-S931N) was attached to the door, and the gyroscope rotation rate was recorded with the Phyphox application at approximately 460 Hz. Two trials were taken, a hard (fast) slam and a gentle (slow) one, corresponding to a large and a small initial spin. Each record consists of an initial push, a free-rotation phase governed only by friction, and a sharp spike as the door meets the frame. Only the free-rotation phase is analysed. The short interval immediately before impact, in which the deceleration increases as air is compressed against the frame, is excluded, because no term in Eq. (1) can produce a growing $|d\omega/dt|$ and the effect is therefore physically distinct. Figure 1 shows the complete traces with the analysed windows shaded.

4.2 Fits to the Six Models

Each model was fitted with `scipy.optimize.curve_fit`. The resulting parameters and fit quality are given in Tables 2 and 3, and the fitted curves are shown in Fig. 2. In the fast slam the angular velocity decreases only from 2.08 to 1.79 rad/s, that is, to $0.86\omega_0$, and as anticipated in Section 3 all six models fit almost equally well, with R^2 between 0.92 and 0.94. The slow slam spans a wider range, from 0.60 to 0.27 rad/s, and the differences between the models become apparent. A nested F -test comparing the dry model with the dry-plus-Newton model gives $F \approx 4 \times 10^3$ ($p < 0.001$), so that a purely linear law is statistically inadequate and the data require curvature.

4.3 Numerical Check of the Analytical Solutions

To confirm the closed-form expressions of Section 2, Eq. (1) was integrated directly with `scipy.integrate.solve_ivp` (RK45) using the fitted coefficients, and the result was compared with the analytic curve (Fig. 3).

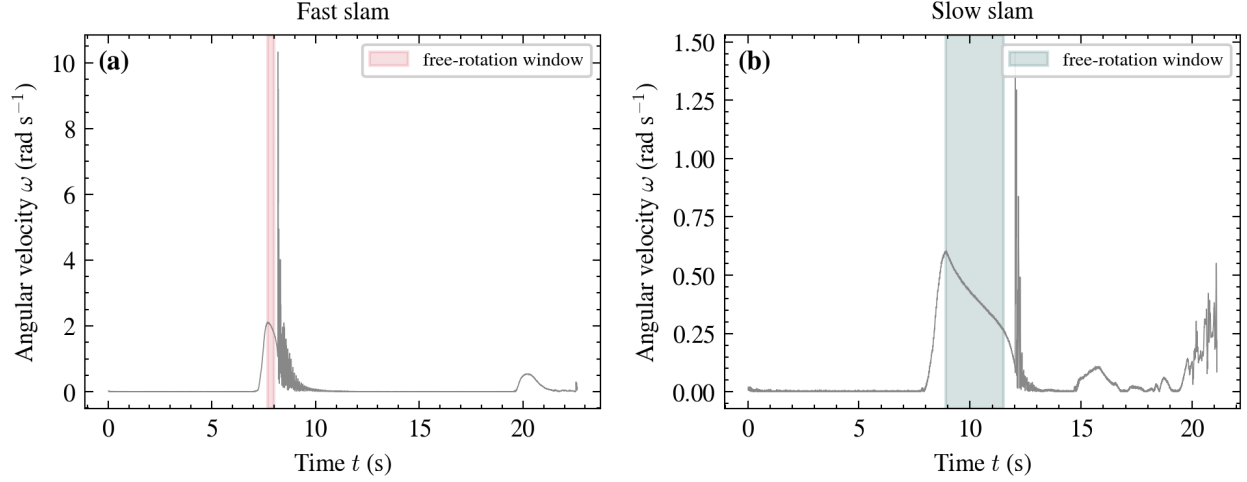


Figure 1: Gyroscope traces for the two slams, with the free-rotation windows shaded. The sharp spikes mark the impact with the frame and are excluded from the fits.

Table 2: Fitting parameters and goodness-of-fit — fast slam.

Model	a/I	b/I	c/I	R^2	SSE	ΔAIC
D	1.088	—	—	0.9441	0.08098	0.0
S	—	0.5432	—	0.9343	0.0952	23.3
N	—	—	0.2706	0.9238	0.1104	44.7
DS	1.088	1.923e-07	—	0.9441	0.08098	2.0
DN	1.088	—	4.58e-18	0.9441	0.08098	2.0
SN	—	0.5432	1.78e-20	0.9343	0.0952	25.3

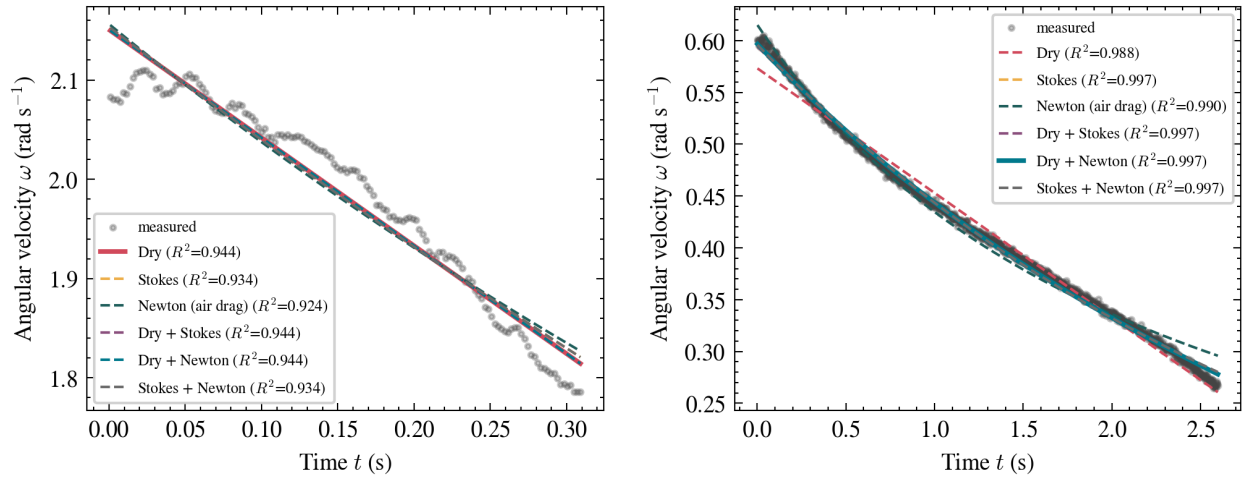


Figure 2: Six-model fits to the free-rotation phase of the fast (left) and slow (right) slams. In the slow case the straight dry-friction line clearly misses the curvature of the data.

The two agree to about 10^{-8} rad/s, which is at the level of numerical round-off and confirms the derivations. The same routine also applies to the general model, for which no simple closed form

Table 3: Fitting parameters and goodness-of-fit — slow slam.

Model	a/I	b/I	c/I	R^2	SSE	ΔAIC
D	0.1203	—	—	0.9879	0.12	1785.3
S	—	0.2911	—	0.9967	0.03321	236.5
N	—	—	0.6752	0.9896	0.103	1601.9
DS	0.0001393	0.2908	—	0.9967	0.03321	238.5
DN	0.05793	—	0.3556	0.9973	0.02725	0.0
SN	—	0.2699	0.0495	0.9967	0.03277	222.4

is available.

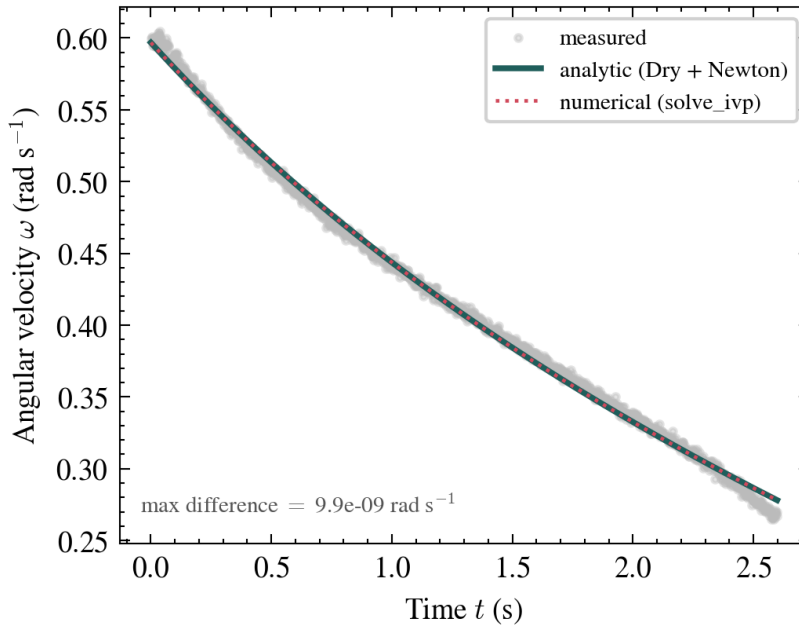


Figure 3: Analytic solution versus direct numerical integration for the slow slam. The curves are indistinguishable (maximum difference $\sim 10^{-8}$ rad/s).

5 Critical Analysis and Commentary

Because goodness of fit alone cannot select among the models, each fitted coefficient is compared with an order-of-magnitude estimate obtained from first principles, following the original paper. For dry friction $a/I = 3\mu g/w$, for Stokes drag $b/I \lesssim 18\pi\eta w/m$, and for Newtonian drag $c/I \lesssim \frac{3}{2}C_D\rho Aw/m$, with $\eta = 1.7 \times 10^{-5}$ Pas, $\rho = 1.3$ kg/m³, and $C_D = 1.5$. The comparison is summarised in Table 4 and Fig. 4.

The fitted Stokes coefficient is of order 0.3 to 0.6 s⁻¹, roughly 10^3 to 10^4 times the estimate of 6×10^{-5} s⁻¹, so that Stokes drag is far too weak to account for the observed deceleration. The dry-friction model appears adequate until the implied sliding coefficient is examined. It is $\mu \approx 0.031$ for the fast slam and $\mu \approx 0.0034$ for the slow one, a factor of nine apart. Since μ is a material constant and cannot depend on the initial spin, the dry model is internally inconsistent; Klein et al. rejected it on the analogous ground that the implied torque was unreasonably large. The Newtonian

Table 4: Order-of-magnitude validity of the fitted friction coefficients (cf. Klein et al.).

Coefficient	Fitted (fast / slow)	Theoretical estimate	Verdict
a/I (Dry)	1.088 / 0.120	$3\mu g/w$, $\mu=0.031/0.0034$	Rejected (μ $9\times$ inconsistent)
b/I (Stokes)	0.543 / 0.291	$\sim 6 \times 10^{-5}$	Rejected (10^3 – $10^4\times$ too large)
c/I (Newton)	0.271 / 0.675	$\lesssim 0.30$	Accepted (same order)

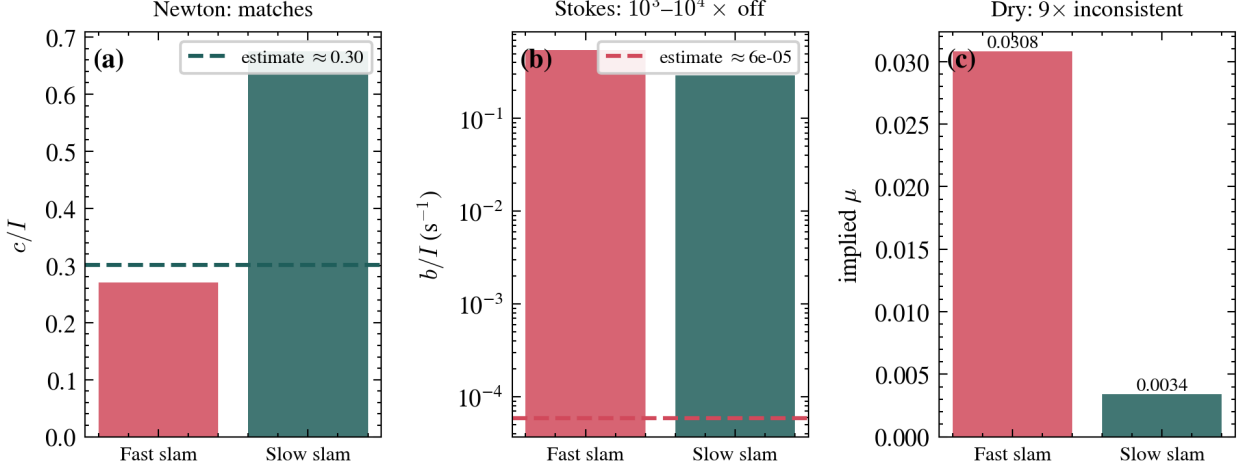


Figure 4: Fitted coefficients against their physical estimates. The Newtonian c/I lands on the predicted value; the Stokes b/I overshoots by three to four orders of magnitude; and the dry-friction coefficient implies two incompatible values of μ .

coefficient is the only one consistent with its estimate. The fitted values $c/I = 0.27$ (fast) and 0.68 (slow) are of the same order as the predicted 0.30 , and the increase of c/I at lower spin reproduces the trend reported in the original paper, which attributed it to the Reynolds-number dependence of C_D .

The limitations of the measurement should also be noted. The fast slam is too short to discriminate the models on its own; the same limitation of the slammed-door method is reported in the original paper, where ω decreases only to about $0.75\omega_0$ before impact. In this shorter and noisier window the unconstrained fit even favours the simplest (dry) model slightly, which is the situation described in Section 3 and is resolved by the physical-validity test. Klein et al. avoided the difficulty with a separate, longer laboratory experiment on a light plate. The natural extension of the present work is to repeat the measurement with the door released from a wider angle at higher spin, so that it decelerates over a longer arc without striking the frame. Further sources of uncertainty include gyroscope noise, which appears as the lower R^2 of the fast window, the uncertainty in r , and the thin-plate approximation for I .

6 Conclusion

A real door was measured with a smartphone gyroscope, the friction models of Klein et al. were reconstructed with their intermediate steps supplied, and the analytic solutions were verified by direct numerical integration. Of the six models, the Stokes and dry-friction terms are excluded on physical grounds, the former by a discrepancy of three to four orders of magnitude in its

coefficient and the latter by a coefficient that does not remain constant between trials, leaving the quadratic Newtonian air-drag term as the only physically consistent description. This agrees with the conclusion of the original paper. Beyond the specific result, the project illustrates a general methodological point, that agreement with the data is not sufficient to establish a model and that the fitted parameters must be confronted with independent physical reasoning.

References

- [1] P. Klein, A. Müller, S. Gröber, A. Molz, and J. Kuhn, “Rotational and frictional dynamics of the slamming of a door,” *American Journal of Physics* **85**(1), 30–37 (2017).
- [2] Team solvE, *physics-door-slam*, analysis code and data, <https://github.com/eeuunn/physics-door-slam>.